BRIEFING

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Workshop on Generative AI & Copyright



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Workshop on Generative AI & Copyright

Abstract

The JURI Committee workshop on generative AI and copyright was held on June 4, 2025. Prof. Dr Lucchi presented a study requested by the Committee on the legal aspects of the phenomenon. The complex relationship between copyright and generative AI from the perspective of technological and economic dimensions was addressed by the accompanying expertise of Dr Axel Brando and Prof. Dr Peukert, respectively.

The workshop included a perspective from the creative sector, as well as a presentation of views from the European Commission.

These proceedings are prepared by the European Parliament's Policy Department for Justice, Civil Liberties and Institutional Affairs for the JURI Committee.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ART	Authorization, remuneration and transparency
BSC	Barcelona Supercomputing Centre
CDSM	The Copyright Directive (Directive 2019/790)
CMO	Collective Management Organization
CULT	European Parliament's Committee on Culture and Education
DMA	Treaty on the Functioning of the European Union
DMCA	The Digital Millennium Copyright Act (US)
DSA	The Digital Services Act (Regulation 2022/2065)
ECR	European Conservatives and Reformists
ECSA	European Composer and Songwriter Alliance
EMMA	European Magazine Media Association
ENPA	European Newspaper Publishers' Association
EPC	European Publishers Council
EPP	European People's Party
EU	European Union
EUIPO	European Union Intellectual Property Office
EVA	European Visual Artists
EWC	European Writers' Council
FEP-FEE	Federation of European Publishers
GESAC	European Authors' Societies

HEC	École des Hautes Études Commerciales, University of Lausanne
HPES	High-Performance Embedded Systems
JURI	European Parliament's Committee on Legal Affairs
LLM	Language learning model
MEP	Member of the European Parliament
SAA	Society of Audiovisual Authors
S&D	The Progressive Alliance of Socialists and Democrats
TDM	Text and data mining
TFEU	Treaty on the Functioning of the European Union

EXECUTIVE SUMMARY

On June 4, 2025, Prof. Dr Nicola Lucchi presented his study on generative AI and copyright to the European Parliament's JURI Committee. His research explored the legal questions of whether the use of copyright-protected content to train generative Ai models is lawful under current EU law, and whether the output of this system qualifies for copyright protection and if so, who owns the rights. The central thesis of this study is that Articles 3 and 4 of the Copyright Directive (CDSM) were intended to support the expressive recombination and large-scale model training practices characteristic of generative AI systems, which go beyond the analytical purposes of text and data mining exceptions. The study also included an examination of Article 53 of the AI Act, ascertaining that the provision should not be read as validating the use of protected content under Article 4 of CDSM. The study calls for the implementation of four structural reforms: (1) a statutory remuneration model for creators, (2) the restoration of prior authorization, (3) clarification on the legal status of generative AI outputs, and (4) an administered output registry with audit and traceability tools.

The discussion was supplemented by the presentation of Dr Axel Brando, a mathematician and computer scientist leading a Multidisciplinary Research Group on Trustworthy AI at the HPES lab of the Barcelona Supercomputing Centre (BSC). He explained how AI "learns" in a fundamentally different way from humans, namely by statistically approximating data points through high-dimensional hypersurfaces, rather than understanding the content itself. This has major implication for copyright and authorship, as it poses the issues of attribution and traceability – a technical challenge with legal implications. The essence of the issue is that models do not store or recall specific inputs but generate outputs based on abstract mathematical structures, making it difficult to determine whether what is produced is new or derived from copyright material. As there is no concrete and specific link between outputs and specific training data, legal questions on infringement of copyright and authorship become complex.

This was followed by an examination of the issue from an economic perspective by Prof. Dr Christian Peukert, Associate Professor of Digitalization, Innovation and Intellectual Property, HEC Lausanne. His presentation highlighted how digital tools have both challenged and expanded creative output. Early efforts to fight piracy failed but licensing platforms proved more effective. An overview of the empirical research literature on the topic shows that without compensation for creators, content quality and contributions decline, as demonstrated through real-life examples. He presented policy options based on both theory (what to do) and empirical evidence (what not to do), with statutory licensing offering a practical solution when rights are unclear. Ultimately, he concluded that copyright must balance innovation with fair compensation, ensuring sustainable data supply while supporting the broad societal benefits of AI.

The workshop also dedicated time for perspectives from the creative sector by giving the chance of organizations representing various artists to present their points of view. The speakers highlighted the current grey area that AI has created regarding authorship, remuneration and transparency by using copious amounts of creative works without proper attribution or subsequent remuneration.

This was followed by a short reflection from the expert speakers, and then the opportunity for Q&A was given to Members of the JURI Committee.

1. PRESENTATION OF THE STUDY ON GENERATIVE AI AND COPYRIGHT – PROF. DR NICOLA LUCCHI



Good morning. Let me start by thanking the JURI Committee and the Policy Department for inviting me to work on this topic that, for an intellectual property scholar, is one of the most intellectually exciting and relevant right now.

This study revolved around the two core legal questions:

1. **Is the use of copyright-protected content to train generative AI models lawful under current EU law?**
2. **Can the output of this system qualify for copyright protection and, if so, who owns the rights?**

These are not hypothetical issues. Generative AI directly challenges three foundational principles of copyright law: *authorship*, *originality*, and *fair remuneration*. And yet, the framework we currently rely on, namely, **Articles 3 and 4 of the Copyright Directive (CDSM), were never intended to accommodate the expressive recombination and large-scale ingestion practices typical of generative AI training. This mismatch is a central concern of the study.**

Let me begin by saying that copyright has always shown remarkable flexibility. It has successfully adapted from the printing press to photography to motion pictures. The moment we are living in now is not uncommon in many ways. It is just history repeating itself. **What makes this moment different, however, is not only the scale and speed of innovation, but the way copyright is being reinterpreted or, in some cases distorted, to accommodate practices that would otherwise fall outside its original scope.**

We do not need to reinvent copyright. We do not need to reshape it to fit the characteristics of a particular technology. What we need is to preserve its internal coherence through a responsible evolution. That means reinterpreting the core copyright rules once again, in light of new technological shifts, as we have done before.

Let us begin with the output side – that is, the content generated by AI systems. This is, in fact, the less problematic aspect. I want to acknowledge here the excellent work already done on this topic, particularly by two studies commissioned by the European Commission and published in 2020 and 2022. I essentially broadly agree with their findings, particularly on the exclusion of fully machine-generated content from protection and the need for legal certainty around AI-assisted authorship.

On this side, European law is clear on **one key point: works generated entirely by machines, without meaningful human creative input, are not eligible for copyright protection, and we believe this exclusion should be maintained**. It reflects the fundamental principle that copyright exists to protect human creativity.

However, this becomes more complex when we consider artificial intelligence-assisted works. Meaning, those that result from a collaboration between human authors and generative tools. Here, we see

considerable fragmentation across Member States. There is neither a clear nor a harmonized standard to determine authorship, originality, and the minimum level of human contribution.

This legal ambiguity affects both authors and markets. It creates risks of reputational harm, of moral rights being undermined, and of cultural homogenization through algorithmically generated content, that floods the creative ecosystem.

The study calls for a clear and harmonized legal framework, one that distinguishes protected works from unprotected ones based on the level of genuine human input. Importantly, we advise against introducing a new *sui generis* right for artificially intelligent-generated content. Such rights would decouple protection from human authorship and distort the logic of copyright by rewarding outputs that are, in essence, a compilation or a combination of pre-existing works.

Now, let us turn to the input side, that is, the training of AI systems. This is where the legal friction is sharpest and where the current copyright framework reveals its most evident limitations. Training generative AI models requires large-scale reproduction of copyrighted content, often without any knowledge or consent of rightsholders.

At first sight, Article 4 of the Copyright Directive appears to permit this, as long as right holders have not opted out. I say "apparently" because, after a thorough and careful review, I found no solid legal foundation supporting this interpretation. I have examined not only the text of the Directive but also its preparatory work, including the Commission's impact assessment and the 2014 study on the legal framework for text and data mining (TDM) commissioned by the European Commission.

All these sources converge on one essential point: Article 4 was introduced as a targeted experimental tool, with the intention to reduce legal ambiguity and promote data-driven innovation, especially for start-ups and small enterprises.

To give a concrete example of this, the impact assessment consistently framed the exception around cases such as bioinformatics, medical research, or textual analysis for knowledge discovery. It does not refer whatsoever to machine learning, neural networks, or algorithmic model training.

Some argue that the Copyright Directive defines text and data mining broadly. However, as we explain in the study, this view overlooks the context and purpose of the Directive. It is clear that Article 4 was never intended to legitimize the systematic ingestion of expressive, copyright-protected content by large commercial AI models.

Moreover, from the outset, Article 4 was meant to be interpreted narrowly – always within the boundaries of the three-step test under EU and international law. To date, no authoritative legal commentary or academic doctrine has offered a convincing rationale for extending this exception to cover generative AI training.

This referral to the Court of Justice is now pending, raising the prospect of a much-needed clarification. Put plainly, the widespread practice of treating Article 4 as a green light for large-scale AI training has no solid legal legitimacy.

Beyond this doctrinal concern, there are some practical issues. Firstly, opt-out mechanisms like robots.txt or metadata are technically unreliable and easily circumvented. Secondly, the notion of lawful uses remains unclear, inconsistently applied, and legally contested. Moreover, most critically, it fails all three conditions: it is no longer a special case; it interferes with normal exploitation; and it unreasonably prejudices the legitimate interests of rightsholders.

To be clear, generative AI does not merely extract data. It internalizes and recombines expression, such as the structure, the style, and the creative identity of each content. This fundamentally alters the legal equation.

Let me also complete the picture by briefly addressing Article 53 of the AI Act. Some have interpreted this provision, which refers to transparency obligations for training datasets, as an implicit confirmation that text and data mining is lawful for AI training. However, this is a misreading.

As we explain in detail in the study, Article 53 is a transparency requirement—not a copyright rule. It does not and cannot modify the scope of European copyright exceptions, nor does it provide any legal basis to legitimize the mass reproduction of protected works.

In other words, Article 53 implies compliance with existing copyright law; it does not and should not be read as validating the use of protected content under Article 4 of the Copyright Directive.

This begs the question of what can we do? First, the study proposes a quick, pragmatic step, namely the creation of a short-term, high-level expert group. This group should serve as a transitional body to develop enforceable technical standards, to pilot field remuneration models, and to assess whether machine-readable opt-outs can be workable in practice, or if the model needs to shift back to something different.

We believe that a high-level expert group could serve as a flexible space for targeted innovation governance. While the Parliament cannot create this group directly, it can request it formally through a resolution, such as a legislative initiative report or opinion under Article 225 TFEU. This approach mirrors past practice.

Nevertheless, we also probably need structural reform. The study recommends four key actions:

1. **A statutory remuneration model to guarantee creators are not excluded from the AI value chain.**
2. **The restoration of prior authorization.**
3. **Clarification of the legal status of artificial intelligence outputs, especially those generated with human involvement.**
4. **An administered output registry with audit and traceability tools to support transparency and compliance.**

In conclusion, generative AI is here to stay, but so is the principle that human creativity matters and deserves protection. **This study does not call for overregulation. It calls for clarity, coherence, and accountability. In addition, it suggests that the European Union has a real opportunity to lead not only in innovation but also in shaping a fair and sustainable creative future.**

2. PRESENTATION OF TECHNOLOGICAL ASPECTS OF GENERATIVE AI IN THE CONTEXT OF COPYRIGHT – DR AXEL BRANDO GUILLAUMES



The goal for this presentation from my side is to present from a technical perspective why we believe that AI can learn in a different way than what we do as humans, and how this can be applied to various aspects.

The main hypothesis of this presentation is that AI models do not learn like humans. Instead, they statistically approximate data points through a high-dimensions hypersurface. New data points are generate based on this learned hypersurface. This fundamental difference has profound implications for authorship and liability.

The idea here is to try to go through different concepts that are important in the mathematical sense and are important to understand that, at the end, what we are learning is a mathematical model using data. **This data can have some copyright issues, but the point is that whenever we have this mathematical model, it is very difficult to ensure that this model is reproducing something that has copyright or not.**

We will start defining, whenever we say "these data points," what are these data points, and the mathematical function that we are learning. Then we will go to the concept of hypersurface or the mesh that the model is learning. We will go inside the concept of generalization. This concept of generalization has different consequences that we will see later on.

2.1. Data Points and Mathematical Function Concept

Whenever we are learning a mathematical model, it does not matter if it is generative or not. The point is that what these machine-learning algorithms are doing is trying to approximate based on different points. You have different points and, for example, each point could be like an image of a dog. In that case, we have two dimensions. Therefore, what we are doing is reducing the dimensionality from the image, namely the number of pixels, to something that is two-dimensional and placing it on a plane. What we are doing with this blue line is trying to approximate, trying to find the model that best approximates all the points. That is what we are doing whenever we say "learning"—we are trying to find the blue line that best approximates the different points that we have here. Once we have this model learned, then different problems appear with what we do after.

2.2. Hypersurface or Mesh Approximation

Let us imagine that now we want to reproduce this process in high-dimensional spaces, not in two. What are we doing? We are doing the same thing – you have points in a space, and we are trying to

approximate these points in some way. The point is that in these high-dimensional spaces, we cannot imagine what it means, because it is something that goes beyond three dimensions.

To give you an example, an image has, as you can imagine, thousands of pixels. Each of these pixels is one of these dimensions. Therefore, we are approximating surfaces that, at the end, have millions or trillions of parameters, which means that we are working in this million, trillion dimensional space. We cannot imagine it. We cannot say, "OK, we have a hyperplane that is in this way, and we have a point here." It is something that we cannot imagine in our minds, and this causes many problems. Consequently, many things appear for which our intuition does not work as it does in reality. Whenever we study the geometry of these high-dimensional spaces, we see many phenomena that are not happening in low dimensions. This makes it very difficult, for instance, to try to detect when this hyperplane is approximating something that is, in some way, generating something new or something that was part of the training set.

2.3. Generalisation Concept

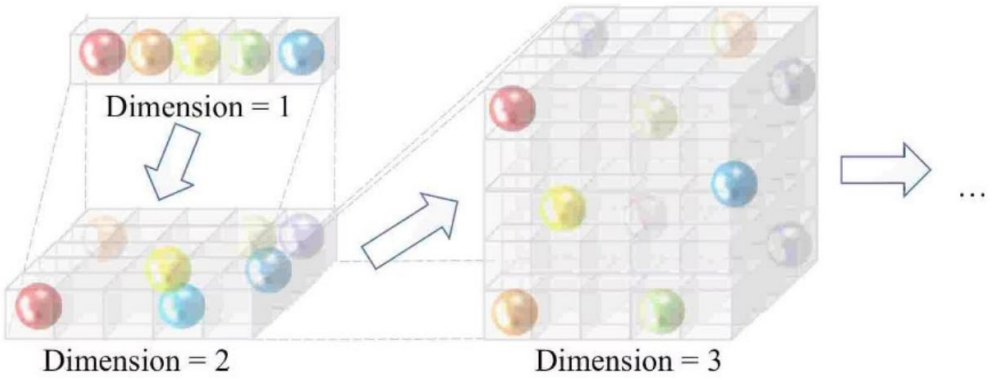
The point of the concept of generalization is that, whenever we have a neural network, this neural network is learning this high-dimensional hypersurface. What we want is that, at the end, it approximates the best possible points that we have but not by memorizing. What we want is that this model tries to generate something that makes sense. For instance, if we are trying to learn a mathematical function that is generating faces of dogs, we do not want to generate only the dogs that we have in the training set. We want a surface that corresponds to the faces of dogs. Then we are able to generate faces of dogs that are not only those we have in the training set.

The point is that making this kind of process is not easy - you can show many examples where we are generating things that really make us believe that what we are generating is something like the face of a dog. Then, many issues appear. One of these issues is the concept of copyright, as you are hearing about right now. The question is then, in what sense is the thing that we are generating something new or not? This happens because, at the end, we are working in a high-dimensional space.

2.4. Dimensionality Phenomena

The concept of dimensionality is about the non-intuitive way these high-dimensional spaces work compared to the single-dimensional spaces.

Whenever you are in a single dimension, an anomaly is something that is not in the range of things that you are working with. In high-dimensional spaces, this is not the case. What happens is that, as you can see here, the balls in the initial part of the graph are more or less spread in space, but they are more or less together at the same time. Whenever we put these same balls in a two-dimensional space, we have many empty spaces. When we put them in three-dimensional space, we have more empty spaces. The point is that finding what is considered an anomaly is very difficult to determine.



This makes our research challenging whenever we are trying to detect whether an image we are generating is new or not. The point is that this is working in thousands or millions of dimensions. Then this phenomenon appears where it is super easy that, only by changing one pixel, we are in an anomaly. Therefore, you need to do the mapping from this concept to the mapping you are working from a legal viewpoint. If you want to say, "OK, this model is in some way making something that is breaking copyright," this issue appears. If I change one pixel, depending on how it is viewed, it could be an anomaly. However, in human terms, it is not an anomaly because we identify the same dog as the original input. Therefore, it likely has a copyright. This needs to be explored.

2.5. Stochastic Parroting

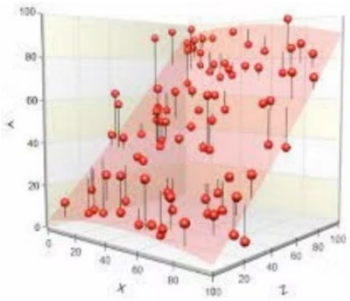
This is a concept that is very related and orthogonal to the previous one. Can we ensure that what we are generating is not doing stochastic parroting? If you go to the literature, you will see that this concept appears in the LLMs. For instance, there are people saying that what an LLM does is storytelling like a stochastic parrot, generating things that imitate what it has seen before. It is something that is not very clear. However, what we see here is that it is true that it is based on things that exist. We cannot easily ensure that what is generated is new. We do not even have any guarantee that it is in fact new.

Given the technical concepts explored so far, we can see that therefore, the concept of authorship and originality becomes more complex in the context of AI. It requires us to mix not only technical but also legal considerations. It is a multidisciplinary problem; therefore, it cannot be dealt with exclusively from a legal or from a technical perspective only. It should be approached from different perspectives.

From the technical side, we have real challenges in trying to detect when something is an anomaly or not. From the legal side, this refers to whether something is new or not. In both perspectives, that remains a challenge that is yet to be addressed.

2.6. Attribution

The idea of stochastic parroting also connects with the concept of attribution. Whenever we observe that we are generating certain points and decide that we want to learn where they come from and then understand if it connects to data containing copyright that was used for training, it is not easy to do so because what we preserve as the output is the hyperplane.



Essentially, whenever we are generating images, is this red hyperplane that you can see here. We are not preserving the points. That makes sense because we are training these models with millions or billions of data points. It does not make any sense that each time we want to generate something, we compare it with these millions or trillions of points, it would be computationally infeasible.

The point is, we are working with this mathematical hyperplane (the red surface) but throughout the process we lose the points. Therefore, we lose the trace of whether the generated output came from something that has copyright or something that is in a new area. This poses a challenge for applying attribution.

This poses the question of why are there elements that make this kind of citing process, and the answer to that is this normally comes from a post-process. Meaning, it is not something that happens while we are generating the output. We generate something first, then determine that certain aspects are close, and then we add those references. However, we do not have any guarantee that these references are really the reason why this hyperplane is approximated in this way.

This poses are huge problem in attribution, and it therefore affects the legal side.

2.7. Conclusion

There are two important aspects from a technical perspective that, in my opinion, directly impact the legal matters being discussed today.

On one side, it is not easy to make these systems able to detect whether what they are generating is novel or not. It is something that will require a lot of research and multidisciplinary work. On the other side, there is the issue of attribution. It is not easy to detect if the point we are generating comes from several different points because, at the end, we are preserving the hyperplane.

3. GENERATIVE AI, COPYRIGHT AND ECONOMICS – PROF. DR CHRISTIAN PEUKERT

I will provide an economic perspective on copyright in the digital age, and my intervention is based on our own research and a recent comprehensive review of the empirical literature. I will also touch on the latest developments in the field.



3.1. Historical Background

In the early days of digitization, around 1999, file sharing emerged as a major copyright infringement issue. The key takeaway from lots of research on file sharing is that fighting online piracy did not really work. Consumers wanted to access music and movies online, and what helped rightsholders recover revenues was licensing digital services like *iTunes* and *Spotify*.

At the same time, digital technology did not just weaken copyright, it also made it easier and cheaper to create content, resulting in a massive increase in the supply of creative works. We have seen about three times more music, movies, TV shows, video games, comic books and, of course, new forms of creative outputs. Much of this new content is user-generated and often involves the reuse of existing works, introducing what you would call a "supply-side copyright challenge". To illustrate the scale of this we can look at *YouTube* as an example, which is probably the largest platform for user-generated content, and hosts over 14 billion videos of which about 50% actually include music. This highlights a big copyright problem.

Let us continue with *YouTube* as an example of a platform with user-generated content. How does *YouTube* handle copyright? Historically, we have seen a shift from pure enforcement, such as takedown notices under systems like the DMCA, to enabling a low-cost, large-scale licensing market for reuse via technology called *ContentID*. This system allows rightsholders to algorithmically detect uses of work and then choose to monetize them. It is a relatively efficient solution that generates substantial revenues for rightsholders. However, it also circumvents traditional institutions like courts, and of course, it carries risks such as fraud and a bias toward economically powerful rightsholders.

3.2. What is Different about Generative AI?

Let us examine what we can learn from this short tour of history for today's primary issue, which is generative AI. I brought you a comparative table here where I just talked about file sharing and user-generated content, but we will focus on generative AI:

	File-sharing	User-Generated-Content	Generative AI
Who loses?	(Most) rightsholders	Possibly rightsholders (but may benefit too)	Possibly rightsholders (but may benefit too)
Who wins?	Consumers	Consumers + New creators	Almost everyone (workers, firms, scientists, creators)
Solution	Licensing (e.g., iTunes & Spotify)	Algorithmic licensing (e.g., ContentID)	?

Who loses here? We do not really know—possibly the rightsholders. This is still being litigated, and we do not have clear economic evidence of harm.

Who wins? The answer is also not 100% clear yet, but it is shaping up that almost everyone benefits—workers, firms, consumers, creators—and, on average, the literature suggests that there is a net positive effect.

Therefore, what is the solution that balances all interests? That is, I guess, what this workshop here wants to explore.

What is interesting in this historical dimension is that here, the copyright problem comes with AI having data as a key input, because data might be subject to copyright protection. Some legal scholars argue that copyright now stands in the way of research, in the way of development of AI. One recent *Science* article maps global differences in copyright law for exceptions such as fair use and text and data mining.

Now, we not only see differences in copyright law across countries, but these differences also correlate with relevant outcomes. So, less restrictive copyright regimes tend to have higher AI output—more publications, open-source code, patents, start-ups, and investment. Copyright alone is, of course, not the only factor, but it is nevertheless an interesting correlation.

If countries with fair use, text and data mining exceptions have more AI innovation, then does this mean that every country should abandon copyright law for data, at least? The answer is clearly probably not, because there is important evidence that without compensating rightsholders, high-quality data might dry up.

3.3. Without Compensation, Will We Have Enough Data Tomorrow?

I would like to share two studies with you today, one on *Unsplash*, which is an image-sharing site, and one on *Stack Overflow*, which is a question-and-answer platform. In both cases, researchers have shown that after training data from these platforms was released for AI, user contributions dropped significantly.

The key takeaway for us is that when contributors are no longer compensated or credited, they have less incentive to upload content. Therefore, giving away data for free may work in the short term, but it could undermine the long-term sustainability of quality input data.

Another policy idea is opt-in or opt-out. Let us look at *LAION*, the largest public image-text dataset that has been used to train foundation models. A study shows that only a tiny share of images in this dataset (and these images that we can interpret as opted-in) are of lower resolution, different content, and are highly concentrated, meaning that 10 websites account for 66% of them.

Similarly, only about 2% of images in this dataset are registered in another dataset, which comes from the market-leading third-party opt-out database. Again, these images differ significantly in technical quality—in resolution and size, as well as content. It is even more concentrated, in the sense that 10 websites account for 93% of them.

The bottom line here is that letting rightsholders decide whether they want to contribute to AI training datasets or not will lead to smaller, less diverse datasets, which might not be enough to sustain AI innovation.

3.4. Policy Options Based on Theory

With all this being said, what are our policy options that economists and legal scholars debate at the moment?

The first option is to do nothing. Essentially, let licensing markets develop, which is already happening. Companies like *Google* and *OpenAI* have signed deals with publishers. If markets do not establish, an option is to let the courts decide but also to account for market power. Of course, very large players control parts of the value chain.

The second option is algorithmic licensing. Think back to my example of YouTube and Content ID—but more advanced. Compensation could be tied to how much a dataset contributes to model output. However, we still face a huge practical problem: data provenance is hard to establish. For example, as I mentioned in the LAION dataset case, for 99.8% of the data points, it is essentially impossible to know who holds the rights or whether there are any rights at all.

This brings me to the third policy option, which is statutory licensing. For example, the recent U.S. Copyright Office report has an entire section on this. Here, the idea is that AI developers can access data without explicit consent but have to compensate rights holders.

From an economic perspective, there is much to like about this option, even though it would be less efficient than a market-based solution. In cases where licensing is infeasible, such as large datasets for foundation models, statutory licensing might be more efficient than doing nothing. Moreover, if we have a compensation pool, we would not need to solve this very large provenance problem, which is that we do not know who the rightsholders actually are.

3.5. Policy Suggestions Based on Evidence

Now, some suggestions that are not only theory, but actually backed by empirical evidence—not on what to do, but what to avoid.

Firstly, we should avoid blanket exceptions without compensation because they erode dynamic incentives and reduce data quality. If creators are not compensated, they will stop contributing, which has bad effects on AI output quality further down the line.

Secondly, we should try to avoid opt-in or opt-out regimes because they create selection bias and this would also violate the AI Act requirement that datasets should be representative and complete. I have shown you evidence that opt-in and opt-out data differs systematically in content, resolution, and source.

My final takeaway for you is the following**: Copyright policy must balance the interests of rightsholders with those of research and development, not only including big firms, but perhaps more importantly, universities and non-profits.**

Most importantly, we must consider the interests of society, because generative AI benefits users throughout the economy. Adoption of generative AI is happening at record speed, twice as fast as the computer or the internet, and there is growing evidence of productivity gains. One very widely cited study says that AI assistants increase worker productivity by 15%, and there are many more studies finding similar effects. These societal benefits need to be taken into account when designing the copyright of the future.

4. PERSPECTIVES FROM THE CREATIVE SECTOR

4.1. Society of Audiovisual Authors (SAA) – Ms Cécile Despringre, Secretary General



I am Cécile Despringre, the Secretary General of the Society of Audiovisual Authors. Thank you to the Legal Affairs Committee for giving us the floor today. The Society of Audiovisual Authors is the European association for collective management organizations of authors of audiovisual works. Our 33 members in 25 countries manage the rights for screenwriters and directors in film, television and multimedia industry.

We have two messages this morning. **First, the text and data mining exception is not suitable for generative AI. Second, collective licensing mechanism is the way to deliver remuneration to authors.**

It was already highlighted by Professor Lucchi: **Article 4 of the DSM directive was not written for generative AI models.** The co-legislators did not have in mind these applications that mimic and substitute creative works. As an exception, the TDM provision must respect the three-step test of the Bern Convention and be interpreted in a restrictive way. It could therefore at best only cover limited elements of the use of protected works by general purpose AI models.

We disagree with the Commission's interpretation of the TDM exception, which they consider as applicable to generative AI. This interpretation forces rightsholders to opt out, and in the meantime, AI developers have scraped the whole Internet, including illegal websites and data sets. We also consider that a registry of opt outs would not help authors, instead it would add formalities that would only reinforce the legal uncertainty and waste time and money. What we need is urgent solutions to incentivize licensing markets.

A licensing market must be based on three principles: authorization, remuneration, and transparency; what many of us call the ART of AI. The Commission must clarify how these principles apply to the using of authors' for AI purposes and push AI companies to seek licensing. The answer to the current misuse of protected works is only found in collective solutions. It is the only way to ensure authors are not left behind, particularly in the audiovisual sector. We need action to foster collective licensing mechanisms. The SAA members are well placed to play a role in negotiating such licenses with AI companies so that audiovisual authors will finally be remunerated.

4.2. Eurocinema – Ms Juliette Prissard, General Delegate

My name is Juliette Prissard and I represent Eurocinema, an organization representing producers' organizations. Thank you very much for giving me the opportunity to express the core message that matters to my members. As representing some rightsholders today, I have three key messages to convey.

The implementation of the AI Act will be key, and we hope that the template will help us to implement our authors' and neighbouring rights. These rights are fundamental and enshrined in the EU Charter of Fundamental Rights. We will be at the AI Office's disposal to find the best way to achieve this goal.

As for the Code of Practice, considering the current drafting of the texts, we would rather, along with many stakeholders, have the copyright section deleted.

The second point concerns the future: how do we exercise our rights in the future, now that we do not have the necessary tools and transparency needed? Can the AI Act put in place a system organizing the circumvention of EU copyright law?

We therefore suggest putting in place two major measures:

1. **A presumption of data mining by generative AI providers, if a judicial body identifies strong indicators of such behaviour. The burden would fall on AI providers to rebut this presumption.**

2. **An injunction procedure addressed to AI providers by the judge to disclose elements of proof.**

We trust that these measures would give rightsholders leverage for the negotiation of future licensing agreements that we are advocating for. Voluntary licensing agreements and collective licensing with all rightsholders. These agreements can be structured to allow producers to allocate remuneration based on contracts with rightsholders and agreements managed by CMOs representing authors, performers, and producers.

We indeed are convinced that only contractual freedom, enhanced with voluntary agreements with CMOs, will give the possibility to rightsholders to get a fair price for the work that they have created, interpreted, and produced.

Finally, we want to emphasize that we have carefully read Alexandra Bensamoun's report on the remuneration of cultural content used by generative AI, presented to the CULT Committee on May 20th. This report is brilliant, and we support it.

4.3. European Composer and Songwriter Alliance (ECSA) – Mr Marc du Moulin, Secretary General

My name is Marc du Moulin. I am the Secretary General of the European Composer and Songwriter Alliance. We represent 58 professional organisations representing 30,000 composers and songwriters across 28 European countries. Thank you for this invitation.

This report is a crucial opportunity to engage in a long-overdue democratic debate on generative AI and its impact on copyright. A debate that never took place when the text and data mining exceptions were adopted five years ago.

Generative AI models already exploited massive amounts of copyrighted works without any consent, transparency, and remuneration for the creators we represent, and none of our members has been able to reserve rights in practice. These models, such as *Suno AI*, would not exist without the work of our members, but now directly competes with them.

We have never seen an exception to copyright so unclear, so widely misused, and causing such devastating impact on music authors. This is not just unfair, it is unacceptable.

Therefore, we urge the Parliament to reassess the applicability of the text mining exception to generative AI and to prioritise consent, transparency, and remuneration.

The AI Act was a step in the right direction, but we have serious concerns about its implementation. At this stage, **both the current draft Code of Practice and the transparency template will not provide enough transparency** to music authors to exercise and enforce their rights. **The AI Act should be implemented with the highest level of transparency and make sure that EU copyright law truly applies to all models, including those based outside Europe.**

If we want to foster an ethical and sustainable licensing market, we must consider effective and enforceable mechanisms to remunerate music authors from the AI-generated output.

Establishing a presumption of use of copyrighted works could play an important role of rebalancing the playing field and encouraging fair licensing agreements. Those agreements should not only be made between large and dominant players with threats to both fair competition and cultural diversity. They should respect the consent of creators and ensure that they can benefit from an appropriate and proportionate remuneration for the exploitation of their works.

4.4. European Authors' Societies (GESAC) – Ms Adriana Moscoso, General Manager

I am Adriana Moscoso del Prado, General Manager of GESAC, representing more than 1.2 million creators from all artistic fields from across the European Union through our 32 member CMOs.

GESAC welcomes this initiative report and strongly believes that it can contribute to properly implementing the existing acquis. The main principles that must guide this work: consent, transparency, and remuneration. Because today, the primary problem we have is the non-

transparency and the non-compliance of generative AI services operating in the EU, and the refusal to remunerate creators.

We believe that the current legal framework, if applied properly, can ensure the necessary remuneration and development of a licensing market. The current practice is far from being satisfactory. For that, we believe **the most important policy considerations** that we would like to raise today, are the following.

Firstly, we need the creation of a functioning licensing market, and for this, we need to require generative AI services to cooperate in good faith with CMOs and rightsholders to either obtain a license or remove unauthorized works. So far, the Code of Practice process has been completely useless for this.

In this regard, **if this first requirement is not met, we need to think about new legal solutions**. For instance, a rebuttable presumption of usage for generative AI services if they are not sufficiently transparent with their procedures.

Secondly, transparency applied across all the generative AI value chain is a meaningful way to allow rightsholders to exercise and enforce their rights. Unfortunately, the draft transparency template falls very short of that so far.

We also need to focus on initiatives, on actions that will allow immediate access to the exercise of rights, instead of new technical, costly, and unworkable ideas, like the creation of a central repository of information, which we reject, or a new dialogue to develop machine-readable opt-outs. We cannot lose more time, and we have solutions today in the market.

In addition, we need an application of EU copyright law to all generative AI services operating in the EU – EU ones and non-EU ones. That would be the only way to avoid unfair commercial competitive advantage from non-EU-based AI companies that circumvent EU law.

Finally, we need to guarantee the respect of the principle of appropriate remuneration of authors and encourage collective management as the most effective way, for both rightsholders and companies, to facilitate licensing and legal security to the market.

We believe that the use of AI-generated output might also need further consideration under this study because of the confidence in the detrimental substitution effect it may have in the market.

4.5. European Visual Artists (EVA) – Ms Laetitia Nguala Masamba, Business Process Manager

Thank you for the opportunity of being here today. My name is Laetitia Nguala Masamba, and I am representing EVA. EVA represents 31 collective management organizations for visual arts who in turn represent up to 170,000 visual artists worldwide.

Firstly, EVA fully endorses the legal assessment expressed by the Spanish and Portuguese governments on the non-applicability of the TDM exception to AI, as also shown in the study by Professors Dornis and Stober.

When it comes to licensing, **EVA members are favourable to licensing, but on certain conditions**. We think that it is important that the remuneration should be sufficient, considering the unique value of artworks. We also think it is important that no harm should come to authors.

Some of our members are already working on licenses, such as in the context of reprography, which extends to AI, where the content is intended only for internal use and not to replace the future commissions of visual artists. **There should also be control on the input and the output**. For instance, restrictions on the creation of copyright-infringing works. **There should also be a choice for the rightsholders or authors whether to license or not, especially in the cases of primary licenses of individual artists.**

However, the biggest issue that we face with licensing is that the popular artworks of high value to AI model providers have already been mined. Many generative AI models can reproduce images in the styles of artists in our members' repertoires, like Dalí, Chagall, Magritte, etc., leaving our members with very few licensing opportunities.

We therefore need, urgently, mechanisms to bring AI model providers to the table to negotiate licenses, perhaps even an obligation to unlearn. Only then can we really regain the power to negotiate the works of our authors.

I would also like to speak about the respect of the copyright acquis. Just to clarify: **visual artists are not against innovation. Visual artists are, in fact, innovative by nature.** However, they want to see the respect of their works and of the copyright acquis. **There must be consent before use of works, and infringements, like for the works that were unlawfully mined before the adoption and implementation of the DSM Directive, must be addressed**. Same for the mining of works from pirated sources that is going on until today.

It must also be clear that AI models used in Europe must comply with EU copyright law, regardless of their country of origin or training. Big Tech and AI providers should not benefit from exceptions and limited liability regimes that disrespect and harm individual authors and artists throughout the EU.

4.6. European Magazine Media Association and European Newspaper Publishers' Association (EMMA/ENPA) – Mr Ilias Konteas, Executive Director

My name is Ilias Konteas, and I am the Executive Director of EMMA and ENPA, who are the two largest trade associations representing the voice of European publishers across Europe.

For today's discussion, I would like to start with two examples that highlight the scale of the challenge:

1. **Google's AI Overviews that significantly impact press publishers by driving zero-click searches, diverting traffic and revenue, and disrupting editorial strategies.**

2. **The proliferation of parasitic AI-generated press websites, the so-called content farms**.

What we want to see, and we hope that this will be reflected also in the end report by MEP Voss**, is a framework that is fair, future-proof, and puts at its core human authorship; prevents the unlawful and unregulated use of protected content by AI; and addresses the displacement of creators and journalists from the market.**

Transparency is core, and transparency should be the focus. Without it, nothing else will work. What we need is full and actionable transparency to ensure accountability and legal compliance in AI development and deployment. We need full disclosure of any use of protected content for any purpose.

I would also like to stress that access to proprietary content without knowledge and consent does not constitute a trade secret under EU law and cannot be used as an excuse to evade transparency and source documentation. There are studies that show the technical feasibility of full source documentation.

We also support the idea of a rebuttable presumption of use by AI companies.

Concerning the opt-out mechanisms, we agree with those who have said that these do not resolve the underlying issue of opacity in AI content usage. They shift the burden of proof and enforcement onto the creators and provide no guarantees that AI companies will respect them.

In such a way, we ask for proactive obligations on AI companies instead of reactive measures by rightsholders.

I leave you with these final words: transparency is the core priority. Without it, nothing else will work. Creators' control and enforcement of their rights must be guaranteed.

Our focus must be on real solutions. AI users cannot use, without permission, content that does not belong to them. It is as simple as that.

Fairness, accountability, and respect for human creativity must guide Europe's digital future.

4.7. European Publishers Council (EPC) – Mr Nikolas Moschakis, Senior Policy Advisor

My name is Nikolas Moschakis. I am Senior Policy Advisor at the European Publishers Council, which is a high-level group of chairmen and CEOs of leading media groups in Europe, representing companies that are active in newspapers, magazines, television, radio, digital marketplaces, journals, e-learning, databases, and books.

The mission of EPC is to advocate for a free, comprehensive, and sustainable media environment that supports high-quality journalism and safeguards the economic viability of the publishing industry. Protecting intellectual property rights is part of our core mission.

What we face today with generative AI tools, some embedded in Big Tech companies, is not just another wave of innovation. It is a fundamental threat to media sustainability and publishers' ability to serve democracy.

A lot has been said already by my colleagues, but I would like to highlight a couple of points. **Mainly, this innovation is based on copyright theft by multibillion-dollar companies that are setting the new rules of the game, while there are dissenting voices, start-ups, and smaller AI companies trying to comply with the law.**

The EU is well equipped. **We need urgently to enforce existing rules such as the Copyright Directive, the DSA, and the DMA. These frameworks must compel AI developers and platforms to be transparent about what content they have used, are using, or intend to use, and to negotiate licenses.**

A sufficiently detailed summary of training data is not optional. It is a legal necessity for publishers to enforce their rights. For this, we find the current template inadequate, and we are waiting to see the new version coming from the EU AI Office. It is the only tool for us to be able to enforce our rights.

Six points to highlight:

1. **Treat the TDM as an exception, which it is, not as a right to train.**
2. **Transparent and meaningful disclosures from AI developers must be required.**
3. **We need a strong and meaningful template to provide the right conditions for licensing, whether for training or grounding or dynamic content injections.**
4. **Reject robots.txt as a means of managing rights. AI companies must respect website terms and conditions.**
5. **Reject the repository of opt-outs. This is unmanageable for rightsholders, as well as expensive and unnecessary.**
6. **Crawling is something else that is out of control. We need to create standards for responsible crawling.**

Finally, demand liability for AI-generated content. No safe harbour for AI content as such.

This is not just a theoretical discussion. It has an impact on our democracy and the ability of publishers to serve it.

4.8. European Writers' Council (EWC) – Ms Nina Georg, President of Honour and Commissioner for Political Affairs

Thank you, Mr. Chair. My name is Nina George, and **I am the writer in the room**. Likewise, I am the Commissioner for Political Affairs of the EWC. The 53 organisations of the European Writers' Council, representing 250,000 writers from 34 countries, **urge political decision-makers to hold accountable AI companies whose profitable generative applications are built on theft**.

We are really, deeply concerned by the exploitation of billions of labour hours and of protected works by non-European AI companies for generative text models, like the illegal usage of 7.5 million books and 81 million scientific papers via the piracy website Library Genesis.

We express our deep dismay that these severe violations are not acknowledged by decision-makers, nor politically debated on the European level. Without our work, our private financial

investment, research, risks, and professional development, no so-called innovative generative models would exist.

Therefore, we demand to hold them accountable, get transparency for every single title used, and make them pay for the non-authorised abuse.

We ask you to preserve Europe's sovereignty and digital independence. Remain committed to AI regulation that focuses on people negatively affected. Revamp the TDM exception – it does not cover generative development at all, as Professor Lucchi rightly pointed out.

Measure the harm done to us, get remedy ready. Do not impose mandatory or statutory licensing on us: enabling to say no to the sell-out of a lifetime work, is the core principle of our position. We know our worth.

We stand firmly, and ask you to do so too, by the principle of ART: authorisation, remuneration, transparency. Thank you, Mr. Chair.

4.9. Federation of European Publishers (FEP-FEE) – Mr Quentin Deschandelliers, Legal Advisor

Quentin Deschandelliers here from the Federation of European Publishers, representing the European book sector.

The book sector is both a user and developer of AI. We are one of the most advanced sectors in the field of out-puts, but we are also a big target. Books, from the side of input, are considered one of the most high-value training data sources for generative AI. On the output side, the market is already flooded with AI-generated fake books that, of course, do not disclose their nature to the public.

What is at stake today is the development of a fair and copyright-compliant AI ecosystem in Europe. The EU will never win a race to the bottom.

The issue in generativity and copyright is not the inadequacy of EU copyright rules; even if we agree that, the TDM exception was never designed for generative AI. The problem is the ecosystem ignoring all rules and infringing copyrights on a massive scale.

What we need is a full duty to be transparent so that rules can finally apply.

Unfortunately, both the Code of Practice and the transparency template are currently dramatically failing this mission. **Transparency is the single most pressing issue—and the prerequisite for everything**.

Training data are not trade secrets. The recipe is a trade secret, not the ingredients.

Without transparency, you have no enforcement, and you have no fair voluntary licensing model. Without transparency, AI companies have every incentive to use illegal sources for their training data. Without transparency, rightsholders cannot properly assess the value of their work and negotiate fair terms for their potential remuneration.

So, what do we need?

Applying the rules – EU rules, copyright rules, AI transparency.

We need transparency on the training data. We need transparency on the data collection crawlers. We need a presumption in courts based on sufficient elements to determine the use for training, as colleagues have suggested. We may also use transparency on the side of platforms providing datasets for training.

What we do not need is, for instance, a registry on opt-outs, which would be costly and not actually provide anything.

5. VIEW FROM THE EUROPEAN COMMISSION – MS EMMANUELLE DU CHALARD, HEAD OF COPYRIGHT UNIT, DIRECTORATE-GENERAL FOR COMMUNICATIONS NETWORKS, CONTENT AND TECHNOLOGY

Emmanuelle Du Chalard thanked the JURI Committee for organizing the workshop, the experts and representatives of the creative sectors for sharing their analysis and their views, and for proposing ideas, options, and recommendations.

She recalled the EU copyright framework, which is based on a number of exclusive rights, that allow the rightsholders to authorize or prohibit the use of their works, as well as exceptions and limitations that facilitate certain types of uses of protected works.

She explained that the TDM exception has become increasingly relevant in the context of AI training, and noted that the co-legislators in the context of the AI Act have also acknowledged the relevance of this exception. A specific feature of this exception is the opt-out that allows rightsholders to reserve their rights and thereby exclude the application of the exception in order to license the use or to prevent the use of their works for TDM.

She mentioned the challenges related to the practical implementation of this exception, and in particular, the opt-out, stressing the progressive uptake of opt-out solutions by rightsholders in different sectors, as illustrated in the study recently published by the EUIPO.

In order to support the practical implementation of the opt-out, the Commission has already launched a study to assess the feasibility and opportunity, of having a registry of TDM opt-outs.

In addition, it intends to further discuss with all relevant stakeholders to identify and agree on the state-of-the-art opt-out solutions.

Emmanuelle Du Chalard also mentioned the AI Act, which includes two copyright-related provisions that will contribute to support the application of EU copyright rules. Discussions on the implementation of these obligations are currently ongoing, particularly in the context of the Code of Practice.

The Commission is working to finalize the template for the summary of training data.

As we are approaching the review of the DSM Directive, the Commission will soon launch a study to support the assessment of the DSM Directive rules, including the TDM exceptions (Articles 3 and 4).

The study will analyse the benefits brought by the new rules, as well as the implementation issues and remaining challenges.

Based on this analysis, and taking into account the need to consider the copyright implications of AI in a comprehensive manner, the Commission will consider the need for new measures to support the development of a licensing market that can, on the one hand, facilitate access to high-quality data for the development of AI in Europe, and, on the other hand, ensure appropriate remuneration for the creators.

6. Q&A

The Chair thanked everyone for their contributions and gave the floor to the experts for a short reflection, before opening the floor to questions from Members.



6.1. Prof. Dr Nicola Lucchi

Well, I perfectly agree with my colleagues who spoke before. **We need to take care to learn from what has happened before with different technologies. We also need to be positive, because, I mean, we *can* learn. Technology disrupts consolidated business models, and even for creators, I think we need to think about how to take advantage of new tools – rather than seeing them only as threats.**

It was the same with the tape recorder, with motion pictures... The cinema was, of course, initially against that technology. However, in the end, that very technology created a new way to be remunerated. The same thing happened with digital technology and the rise of platforms.

Therefore, I am pretty sure that we can take advantage of this technology too. Of course, **now we are in a grey zone, and everything appears to be complex, which creates legal uncertainty and concern, especially for rightsholders. . Nevertheless, I am positive, because it is just history repeating itself. As I said, copyright was able to adapt itself to different situations, always maintaining its core principles, and that is all we need.**

As I discuss in the study, social media platforms, just like AI models, depend on content provided by users and operate through licensing. Without prior authorisation or user consent, they cannot operate legally. In this sense, both are content-dependent infrastructures: empty boxes without user input. That same logic can be applied to generative AI. We already see monetisation models on platforms like *TikTok* that show how usage data and structured licensing, even if not perfect, can work in practice.

Obviously, **many AI companies are currently against everything that might complicate their operations including licensing obligations or transparency requirements. Yet we already have clear examples of licensing systems that can work**. These models can be adapted to the training and output context of generative AI, demonstrating that viable solutions already exist. I am confident that we will find a fair solution for both involved parties – one that ensures innovation continues, but with legal certain and fair remuneration for creators

6.2. Dr Axel Brando

I fully agree with Professor Lucchi's remarks. From a technical perspective, the issues under discussion, particularly those concerning **novelty** and **attribution**, remain unsolved challenges.

As previously mentioned, determining whether a generated output constitutes something *genuinely new* is far from straightforward. Generative models are designed to produce plausible outputs, but assessing their **originality** in relation to the training data is not a trivial task. Similarly, **attribution**, understanding whether a particular output is derived from one or multiple training sources, is technically difficult.

This is why the initiatives being discussed today regarding the attribution and authorship of AI generated content are not only valid, they are essential. However, we must recognise that from the technical side, the tools required to support these initiatives with precision are still under development.

It is important, therefore, to proceed with caution. We are, in effect, placing vast amounts of content into highly complex systems, and then expecting them to return outputs that we can interpret, control, or regulate. At present, we lack the necessary technical infrastructure to fully understand or monitor how these outputs are generated.

This is a **critical area for future interdisciplinary work**, and one in which collaboration between legal scholars, technical experts, and policymakers will be vital.

6.3. Prof. Dr Peukert

I just want to reiterate my earlier point about taking society at large into consideration.

Of course, I understand the concerns of rightsholders; many valid points were raised today. For lawmakers, I think it is really important to remember that AI technologies have benefits for many sectors of the economy, and those benefits should also be taken into account.

Especially when we think beyond copyright. For instance, we should be considering the impact of biased versus unbiased datasets. Sometimes, AI models behave in ways that, from a democratic perspective, we might not want due to the datasets they are trained on.

This is something that needs to be addressed, and the best way forward is to ensure that copyright does not unintentionally get in the way of building unbiased AI. That could happen, for example, if opt-in or opt-out mechanisms lead to a selection bias problem in datasets.

6.4. Questions from MEPs

6.4.1. MEP Axel Voss (EPP) – Rapporteur

I have a huge number of questions. Unfortunately, I am going to have to choose just a few.First, I would like to thank the experts and the stakeholders for their input. I would also like to thank the Policy

Department for making this possible, and the Secretariat for giving us this opportunity here at the European Parliament.

Fundamentally, from what I have heard, there is a feeling that including Article 4 in the AI Act is a kind of breach of copyright law, that it is not appropriate. That is because six years ago, our interpretation of what TDM meant was completely different. It does not belong in the AI Act because it now has a completely different interpretation.

The question therefore is how can we correct this situation?

I believe we need to correct it, with regard to interpretation. However, politically speaking, our problem is that, on the one hand, we do not want to place obstacles in the path of AI, because it is so important for future development. On the other hand, we want creators to have their existing rights respected and we cannot deny them that protection.

This rule in the AI Act breaches international copyright law, as far as I understand.

Do we really need to look at input data and output data differently? Or can we say that we license first in both cases?

Then there is the question of what we can do to stem the tide. I think we have to make an offer to AI developers so that licensing becomes simpler. If transparency helps achieve that, then that is what we should pursue. We need to think about how we can simplify the system.

So my question to the experts is how can we avoid facing the same questions in five or six years? How can we create a system that protects copyright in the long term?

Those are only a few of my questions, but it looks like I have run out of time, so I will have to ask my other questions at a later stage.

6.4.2. MEP Laurence Farreng (Renew) – Shadow Rapporteur

Thank you very much for this very interesting study. While I have not read all 100-odd pages and I have not asked *ChatGPT* for a summary, I have followed your explanations with great interest.

Thank you to the experts. We have already had a chance to discuss these issues together.

First, there is a sense of urgency, and I am very pleased to see Axel Voss's report, I will be able to contribute to it. We are facing the issue of a growing values gap, and it is getting worse and worse. We need to deal with the problem of fraudulent use of data and disrespect for copyright.

Even if these are considered temporary breaches, it seems that major AI models feel no obligation to negotiate with rightsholders, they just scrape up everything. Therefore, I hope that work on the AI agency can be sped up. Commissioner VIrkkunen really needs to get a hold on this subject and take action.

Trust is only possible if there is full transparency around the use of works by AI.

Generative AI systems are very complicated and it can be difficult to introduce traceability measures, even though they are needed.

Therefore, my question is what kinds of mechanisms could be used? By whom and who would bear the cost? If we are going to use expensive technologies like block chain, this becomes an important question.

I also have another question related to the possibility of excluding open-source data from the scope of regulation in order to avoid limiting the competitiveness of some of our companies.

What kind of thresholds should apply? For example, should it be based on the number of users? How can we calibrate this properly?

Then, there is the issue of business secrets. If there is one subject I really want to highlight, it is this – a fundamental issue for the EU.

Can the EUIPO platform ensure confidentiality in negotiations between developers and authors?

A final question regarding the remuneration for authors. **There will be monetary transactions involved, but what about the moral rights of authors over their works?**

In France, moral rights are inalienable and unlimited in time, meaning an author can always oppose any use of their work that damages their personal honour.

6.4.3. MEP David Cormand (Greens/EFA) – Shadow Rapporteur

Thank you very much for the reports and the summaries. I do have some questions. Not because I did not understand, but I think maybe there were some nuances here.

AI is being presented as a simple technological development. In cinema, for example, we moved on to talking films, colour films, and special effects. However, I do not think we are in a situation like that at all with AI. This is a structural change in the business model for culture, with a product that, in fact, exploits creative works.

That takes us far beyond the scope of just technological change – it is a full break.

Concerning business secrets – that really is just a fig leaf, is it not? A way to steal from creators.

A point on what Dr Brando said – this is not the first time I have heard the claim that this is a "technological problem" when it comes to traceability. However, I have a political problem with that. We cannot frame this as a technological problem when actually we are trying to ensure rights are enforced.

If technology is used to undermine people's rights, that is a political issue. We cannot have Chinese walls and say, "Oh well, it's a tech problem, not our business." No. That is unacceptable.

I also thought I heard somewhere that on one side we have authors' interests, and on the other, the general interest of society. As if, authors' rights are in opposition to the general interest of society.

That is as if saying a baker's right to be paid for their bread goes against society's interests. No – workers' rights are society's rights too. Authors' rights are no different.

I would really like to hear more on this idea that the two are somehow opposed.

6.4.4. MEP Angelika Niebler (EPP)

I would like to thank everyone who has spoken it has been very enlightening. I have three main points.

Firstly, what is Europe? It is our culture, our creatives. It is important for all of us here to think about copyright in the era of generative AI, and the protection of rights.

What I have learned from the input so far today is that: (1) the TDM exception is not enough, (2) opt-outs are not working, (3) the licensing market is not working either, and (4) it is important to create transparency.

Now, my questions.

Firstly, transparency is key. If we do not know what is being used, we do not know how to license or remunerate it. **Therefore, I would like to ask our stakeholders and experts what exactly is needed in terms of transparency?**

We want AI to develop in Europe but when I hear "transparency" I think of reporting and disclosure, all of that is quite bureaucratic.

When you talk to start-ups in AI, they tell you it is too heavy. Larger companies are talking about moving out of Europe and going to the UK instead. In such a way, we need to protect rightsholders but also strike a balance.

Then, there is the licensing market. **Should I conclude that statutory licensing is a solution? It has been mentioned as a possibility and I would like to hear more about that.**

Lastly, technical solutions have been mentioned and I would be really interested in hearing what is being done in other parts of the world on this issue.

6.4.5. MEP Maravillas Abadia Jover (EPP)

The presentation of this study has arrived at a crucial point in time. We have to find how we can protect the intellectual property rights of our creators whilst encouraging the development of European industry of AI and ensuring it is competitive. The study makes it clear that the current exception on TDM does not provide legal certainty or enough transparency to guarantee the protection of rightsholders' rights.

There are principles in the study, namely transparency, proportionality, and consistency through the system. We also need to make progress in implementing a fair remuneration system for authors and ensuring that they have the necessary opt out tools so they can ensure that their rights are protected without there being any unnecessary obstacles to development.

This is vital because generative AI will need to have in fact standardized transparency registers of their sources of data so that there can be periodic checks without jeopardizing the health of the industry or the rights of authors. We need this combined approach with the opt outs, traceability of data and fair remuneration systems. If we have that, then the EU can compete and act as a leader when it comes to IPR and AI.

6.4.6. MEP Daniel Buda (EPP)

I just have a couple of considerations on this subject. Mr. Peukert said that for some economic sectors, AI has economic advantages, but we cannot get those benefits if there is no protection for the copyright holders. We simply cannot go into a shop and take something off the shelf without paying for what we take, even if it has economic advantages for us to just take it off the shelf.

We have to protect the rightsholders. Yes, AI is important, but they need sources to work with, and if human beings have created those sources, we need some kind of European mechanism to protect these people and their works. At the end of the day, we will see our creative industries simply dying out.

AI needs to use these sources, but if you use something, you have to pay for it. There needs to be remuneration. You cannot just develop a technology and not pay people for the inputs. If you use somebody else's creative production then you have to pay them for it. These economic advantages cannot simply be obtained at the expense of others.

We have to convert that into legislation now. That is part of our efforts to protect copyright, so we need the right mechanisms to be put in place. AI brings out economic advantages, yes, but we have to pay for what we use.

6.4.7. MEP Mario Mantovani (ECR)

I think it is important to support innovation, but we must not put the brakes on it too much but that cannot happen at the expense of jeopardizing creators' rights. We have to ensure that we properly protect them.

There are areas where there is room for improvement, such as the need to ensure that we have tools to identify content that has been generated by AI, the development of key mechanisms when it comes to opt-outs, and ensuring obligations for transparency are complied with for generative AI systems, particularly regarding the composition and origin of the training data that they have used, thus ensuring that copyright is respected.

Then, the promotion of licensing models and fair remuneration for rightsholders.

An important point is whether we should be protecting AI-generated output if there is no significant human contribution. In the largest digital markets in the world, the US and the EU, content produced only by AI is not subject to copyright protection.

I would like to quote something from Pope Francis: "I believe AI does not have its own reality in a creative or original sense of the term. Its creativity is derived based on models and data provided by humans. Genuine creativeness is still a human prerogative."

I must admit, I am wondering about that – in a world where AI and its developers are, on their own, becoming the holders of the rights that were worthy of protection. In fact, it may be human creativity that is running the real risk, whether in the audiovisual industry, the pharmaceutical industry, the media, or so many other sectors.

Therefore, I think that AI should remain a tool in the hands of humans, rather than having its own right of protection.

6.4.8. MEP Tiemo Wölken (S&D)

I think the presentations we saw were very interesting, and I think it is very obvious that we all have to think and explore new possibilities to strike a system that, in the end, works for everything.

I think that Prof. Peukert rightly said that, in the end, we have to find a system that works for the whole economy and for society. I hope that we will be able to find constructive solutions for this problem. Otherwise, I think we will repeat discussions we have had many times before.

I think we are looking at two dimensions here. **On the one hand, there is the question of what did AI companies do until now? They most likely made many mistakes deliberately, and they may have been in breach of laws. This needs to have consequences**.

The second dimension is how do we adapt our systems to this new normal? This is something I think we cannot answer with old recipes. We have to endeavour a bit of "fusion kitchen" here to find solutions that will work.

The Chair concluded MEP interventions and gave the floor back to the experts.

6.5. Answers by Prof. Dr Nicola Lucchi

Thank you for all these questions. I think I will leave some of the questions for the other colleagues. Before all these questions, I heard that there is one that comes first. We need to clarify what the black box is, how it works, because without knowing these technical aspects, we are not able to regulate anything.

In the paper, I think I replied to almost all of these questions, but in it, I make a parallel that can simplify or identify the core question, that is the parallel with food safety.

Would we accept having food on the market without knowing the ingredients? Nope. We know we do not know the ingredients, we do not know if there is a real copyright infringement or not. Therefore, we need to know this before applying something that makes sense. Otherwise, we are regulating something without knowing the context.

I think this is the aspect that we need to clarify before going further. Otherwise, we probably are regulating something that in a couple of years will be different from the technical perspective.

Then, of course, Article 4 is problematic because there is no other exception. In our system, copyright law works as an opt-in system. Otherwise, we are creating a compulsory license and so, it does not work.

I am very curious to know the study regarding how to transform the opt-out into something that can really work. I talked with different experts in the sector, and of course, we can make a picture of how it works now, the crawling system, but soon there might be a different technology that is able to collect information from the Internet, as well as information from creative material. We are not able to adapt the opt-out system to the new technology.

Another element that we need to consider is that the majority of the AI corporations are not taking care of Article 4. They are doing something more practical. They are concluding contract licenses with the big content creators.

The problem here is how to protect even smaller creators. Of course, access to Springer or other big corporations are able to arrive at fair compensation through licenses and even start court cases in order to arrive at a settlement agreement, but what about small creators? They do not have the capacity now even to protect their rights let alone to be compensated.

6.6. Answers by Dr Axel Brando

MEP Axel Voss asked me something that I think goes directly into two important points here. On one side, he asked how to avoid putting obstacles to AI, but what we can do now. In the presentation I did, I highlighted two terms and that was done on purpose. I think that research should go on the side of novelty – how we ensure or try to have techniques to detect when something that is generated is novel with respect to the generated output.

When are we generating something that is simply a minimal change from what we saw in the past, and when is it something that is absolutely new? I think this is something that, in technical terms, is not solved. I am not an expert on law but the point is that once we have this, we can apply techniques that we are applying to humans to detect whenever something is new and then determine that this is novel. That could be one part.

The other part is attribution. This is something for which we do not yet have a solution in technical terms; how to determine, when we are predicting something with next-generation AI models, whether it came from one source or another. Given that we are training, as we said, with millions or billions of data points, we are approximating this hyperplane, and we do not have any guarantees that the point we are generating is something very close to what exists or something that comes from the generalization process, as I said in the presentation.

The important point here is that these two different concepts should be addressed in some way. We will have better and better techniques on this, as it sure is a very relevant research topic. Nevertheless, we should have at least some kind of techniques for this, in order to try to apply the same kind of ideas that we are applying to humans to detect whenever something is novel or to cite sources for the things we are generating.

Another question that Axel asked in the technical aspect was whether we should look differently at input data and output data. I think that, for this, in mathematical terms it is the same, I do not know how this is determined in legal terms. For me, it does not matter if it is something that comes from a training set or something that I am generating. For me, it is about pointing to this hyperplane.

I think that perhaps we should go in that direction also, in terms of not trying to differentiate between something that is generated or not, because I think this is really another challenge.

If we focus our research on trying to detect if something is generated by AI and not made by a human, I see that it is an important topic. The point is that if we focus on this, we are not focusing on the two concepts I mentioned before, which is whether something is novel and whether something has attribution, that it comes from some source. I think that we are going down another path, and there is a lot of work to do on this.

In fact, MEP Voss also asked what would happen in 5–6 years. I think the problem is now. We trained models with data that existed in the real world and was accessible. Now we have these hyperplanes. For these hyperplanes, we do not have any guarantees that they are generating something that is exactly what exists or not. It is something we need to take care of, because I think we should be careful. This is not something that is only legal, there is a technical side and we need to consider it.

Finally, there was a question about the traceability concept. That is completely aligned with the two concepts I mentioned before: novelty and attribution. I think that traceability has many goals inside, but specifically, there are these two. How we detect whether something we are generating is far away from what exists already, versus determining its novelty, depends on understanding the sources.

I think we should be careful about this, because it is something that is not solved yet.

6.7. Answers by Prof. Dr Christian Peukert

To be succinct given the time restrictions, from an economic perspective I would like to emphasize the differentiation between input and output data. It is really important to make that differentiation because there are different players with different economic incentives involved.

Today's meeting shows that, essentially, most of the stakeholders are coming from the input side. If you had other stakeholders that were more focused on the output side, then you would hear different arguments. So, keeping that differentiation is important.

In terms of transparency, there is also a distinction to be made between input and output data. For copyright, I think input data is where transparency matters and that is what has been discussed today. On the output side, I do not think it is so much a copyright question, but rather a question of whether something is AI-generated. For some instances in society, we might want to know for sure whether something is AI-generated or not.

Then, in terms of technical solutions blockchain was mentioned. I am not so sure whether we need to keep inventing more complex and over-complex technical solutions and bet too much on technology. I think you can just design systems that work. For example, some European countries have exemptions in copyright law for private copying that are essentially like a compulsory license, or a statutory license, where copyright essentially does not apply, but then the rightsholders are still compensated. I think a model based on that, especially in the AI context, would be worth exploring.

This is not without economic downsides, in the sense that it will create inefficiencies. Given that the problem we are trying to solve is so wicked, even an economist is happy to live with some inefficiencies.

Then I wanted to correct the record if there was some misunderstanding on my last slide, where I showed the trade-offs between finding a balance for rightsholders as well as AI researchers, or AI firms, and general society. It is not opposed. These are not things where the economic advantages are entirely opposite to the interests of rightsholders. It is quite the opposite.

Of course, there is also general society's interest in cultural production and in culture itself. That makes it such a difficult problem, which is why I really appreciate that this whole discussion is happening. While I am yet to look into it further, it seems to me that a statutory compulsory license system would be worth exploring in great detail. Again, without any opt-outs, because the opt-outs will just create unrepresentative datasets.

7. ANNEX

7.1. Biographies of experts



Prof. Dr Nicola Lucchi

Nicola Lucchi is Serra Hunter Professor of Law at Universitat Pompeu Fabra in Barcelona, where he focuses on the legal implications of digital technologies, with particular emphasis on intellectual property, AI regulation, and comparative information law. His research adopts a policy-oriented and cross-jurisdictional approach to explore how legal systems respond to technological innovation and algorithmic change. Before moving to Spain, he held academic positions in Italy and Sweden, and has also held research appointments at institutions including UC Berkeley, NYU, Stanford, the Max Planck Institute in Munich, and the Université catholique de Louvain as an EU Marie Curie Fellow. He is an active member of several academic associations.



Dr Axel Brando

Axel Brando is a Mathematician and Computer Scientist leading a Multidisciplinary Researh Group on Trustworthy AI at the HPES lab of the Barcelona Supercomputing Center (BSC). His work focuses on designing robust, explainable, and safe AI systems for high-stakes domains by integrating technical, legal, and philosophical perspectives. He previously completed an industrial Ph.D. with BBVA and the University of Barcelona, contributing foundational research in uncertainty modeling using Deep Learning, with results published in top-tier venues such as NeurIPS, UAI and AISTATS. This research laid the groundwork for the current Trustworthy AI line, coordinating a growing team of researchers and PhD students and engineers. His group's work spans autonomous driving, aviation, finance, pharmacology, legal compliance in AI systems, and fake news detection, through European projects, industrial collaborations, and standardization efforts. Under his leadership, the group promotes a rigorous, multidisciplinary approach to ethical, robust, and legally aligned AI.



Prof. Dr Christian Peukert

Christian Peukert is an Associate Professor of Digitalization, Innovation and Intellectual Property at the Faculty of Business and Economics at University of Lausanne (HEC Lausanne) and Director of the PhD Program in Management. He is interested in how digitalization affects consumers, firms and markets, and how this shapes incentives to innovate. He was previously a senior research associate at the University of Zurich and ETH Zurich (Switzerland), as well as an Assistant Professor of Information Systems and Management at Católica-Lisbon School of Business and Economics (Portugal). He holds a PhD in Management and Economics from the University of Munich (LMU).

7.2. Presentations

7.2.1. Presentation of Prof. Dr Nicola Lucchi





Introduction

1) Copyright has historically adapted to technological shifts — it's not fragile.
2) What's needed is preservation through principled evolution, not distortion.
3) The problem isn't the technology, but how the law is being twisted to serve it (e.g. misreading of the TDM exception)

Output Side: Protection, Authorship & Liability

Highlights:

- Fully machine-generated content is **unprotected** under EU law.
- AI-assisted works need **harmonised protection criteria**.
- Risks: displacement of human creativity, moral rights violations, and cultural homogenisation.

Note:

➤ *Maintaining the exclusion of AI-only outputs is recommended to preserve the human-centred rationale of copyright.*

Legal Gaps in AI Training (Input Side)

Key Issues:

- AI training involves large-scale reproduction—often without consent.
- Article 4 CDSM allows use unless rightsholders opt out, but:
  - Opt-out tools (robots.txt, metadata) are ineffective.
  - Legal uncertainty around "lawful access."
- Article 4 likely fails the **three-step test** under EU law.
- "Generative AI internalises expression—not just data."

Recommandations

Main Proposals (1):

☞ **Create a six-month High-Level Expert Group (HLEG)** to:

- Develop enforceable technical standards,
- Pilot remuneration models,
- Assess feasibility (or not) of machine-readable opt-out tools.

Recommandations

Main Proposals (2):

- Statutory remuneration model.
- Restore prior authorisation for GenAI training.
- Clarify the legal status of AI outputs.
- Harmonised opt-out registry and audit tools.

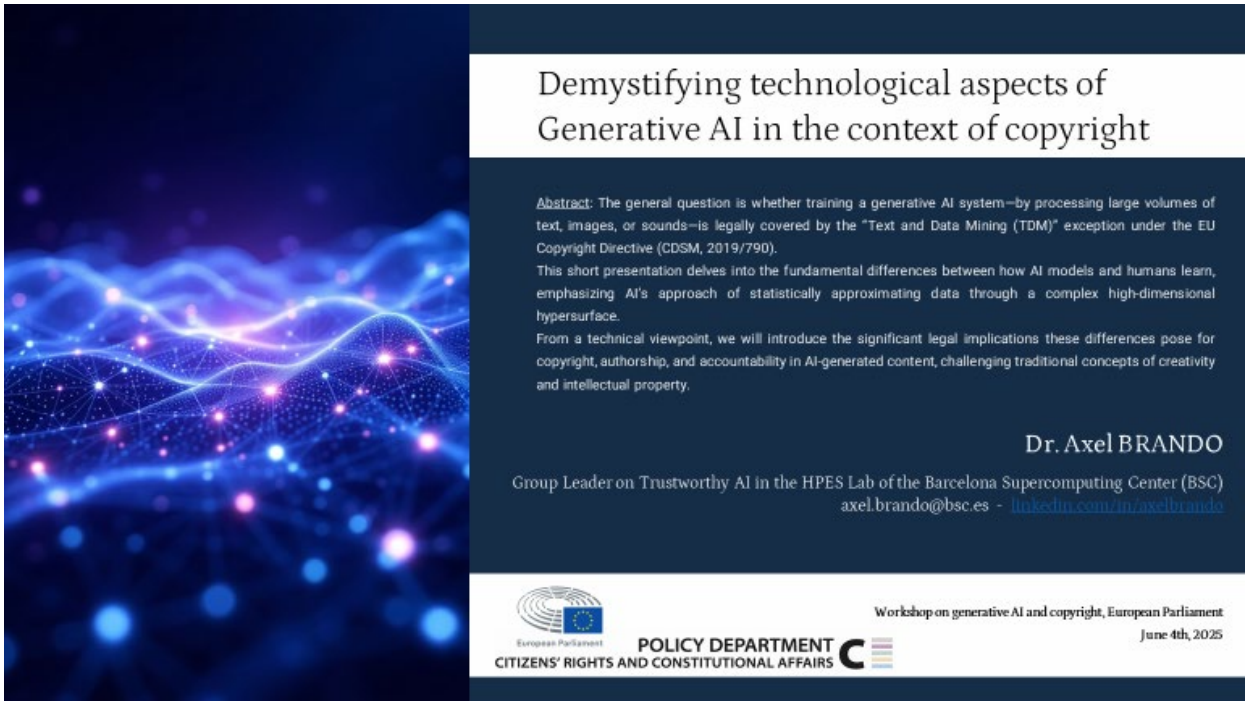
Presentation by

Prof. Dr. Nicola LUCCHI
University Pompeu Fabra, Barcelona, Spain

Policy Department for Justice, Civil Liberties and Institutional Affairs

Responsible Administrator:
Mariusz MACIEJEWSKI
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7.2.2. Presentation of DrAxel Brando



Hypothesis / Main Claim

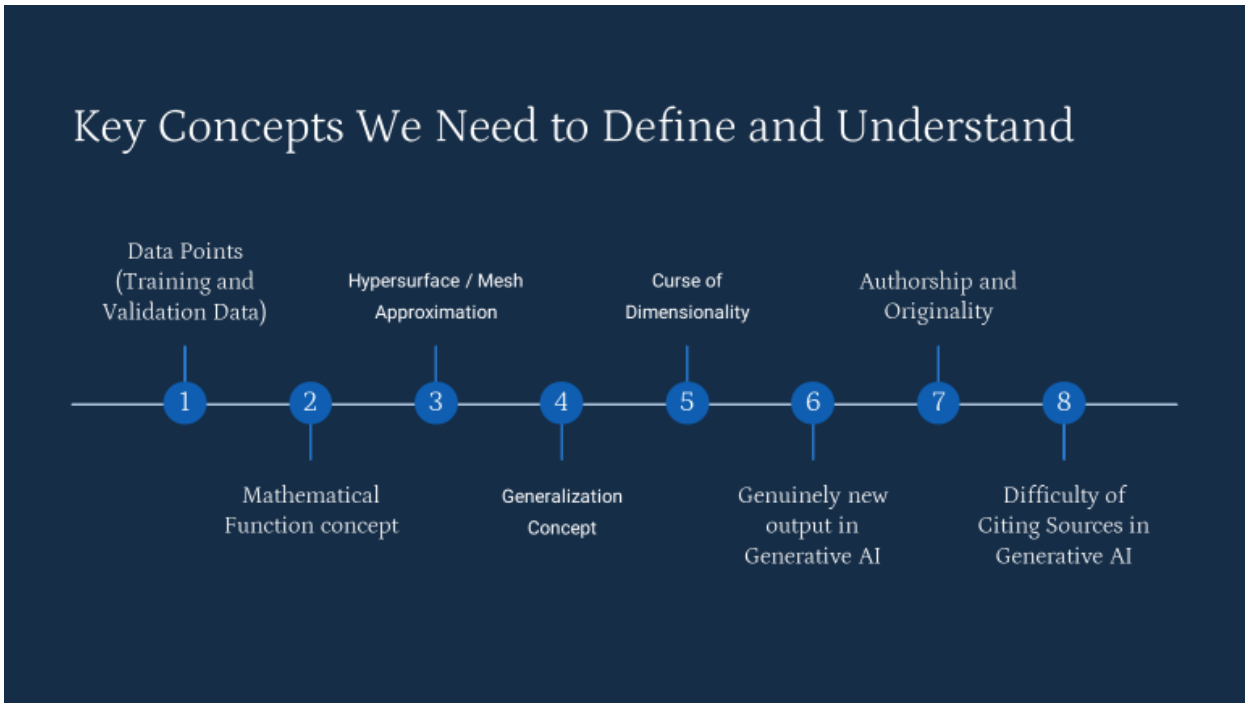
"AI models do not learn like humans; instead, they statistically approximate data points through a high-dimensional hypersurface. New data points are generated based on this learned hypersurface. This fundamental difference has profound implications for authorship and traceability."

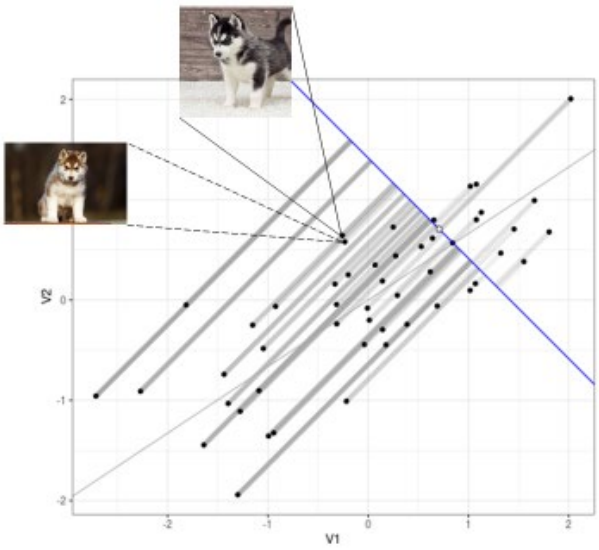
Hypothesis / Main Claim

"AI models do not learn like humans; instead, they statistically approximate data points through a high-dimensional hypersurface. New data points are generated based on this learned hypersurface. This fundamental difference has profound implications for authorship and traceability."

8 Key concepts to explore in this presentation:

Data Points, Mathematical Function, Hypersurface, Generalization, Curse of Dimensionality, Genuinely new outputs, Authorship, Citing.





Mathematical Function: A Simple Starting Point

Let's begin with a simple case:

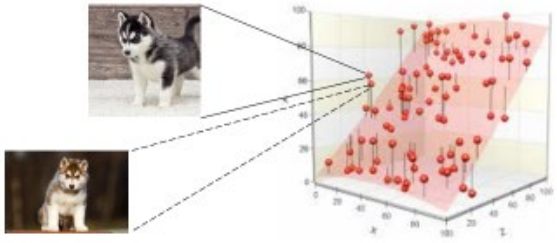
- Imagine we have a **set of data points**: real, tangible instances (texts, images, music).
- **Goal**: Construct a function that approximates these points — rather than memorizing them — enabling the generation of new outputs.
- In basic machine learning:
  - The model approximates these points using a mathematical function.
  - This function can **interpolate** (generate new points within the known range) and **extrapolate** (generate beyond it).

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Key Concepts We Need to Define and Understand

1. Data Points (Training and Validation Data)
2. Mathematical Function concept
3. Hypersurface / Mesh Approximation
4. Generalization Concept
5. Curse of Dimensionality
6. Genuinely new output in Generative AI
7. Authorship and Originality
8. Difficulty of Citing Sources in Generative AI

Hypersurfaces: Moving Beyond 2D-3D Functions

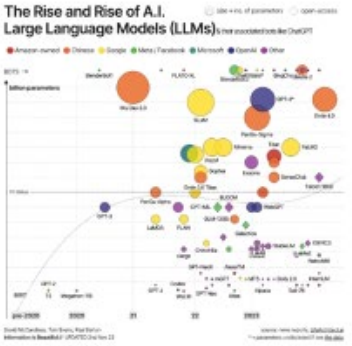


In real-world AI models:

- We operate not in *2D* or *3D*, but in **hundreds, thousands, millions or billion of dimensions**.
- The **hypersurface** is an abstract, high-dimensional structure that encodes the statistical relationships among data points.
- Technically, the model creates a (i.e. *N-1 D*) **mesh** – a surface woven through these vast, high-dimensional (i.e. *N D*) data space.

But critically:

- The training process is **stochastic**:
  - Data is processed in random batches.
  - The model's parameters are updated iteratively.
- **Once the hypersurface is trained, the original training data points are discarded** – it would be impractical (in terms of memory, computation) to compare with all of them.



Thus:

We lose the explicit link between a generated output and specific original data points.

There is no registry or index of the data points that shaped a particular generation.

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4

Key Concepts We Need to Define and Understand





Understanding Generalization in Generative AI

1 **Beyond Memorization**

Generalization allows models to create outputs not seen during training that remain coherent and plausible

2 **Distribution Learning**

This works because the model approximates the underlying distribution of the data—not the data itself

3 **Legal Concern**

Despite generalization, the hypersurface is constructed using protected content, raising critical copyright concerns.

Steps for the generative AI training (see Section 2.1.1 of the Study):

Ingestion

Massive datasets of expressive works (e.g. books, images, songs) are copied and stored—often without consent.

Pre-processing

Data is normalized, cleaned, and tokenized—steps that often involve temporary or intermediate reproductions.

Model Training

The AI builds internal representations of patterns found in the original expressions. These are statistical but depend directly on access to the protected content.

Generation

New outputs are produced by sampling from learned distributions, which are shaped by the structure and style of the original works.

"This process is not merely analytical—as with lawful TDM—but rather reproductive in nature. **The model does not just "learn" facts or trends; it internalizes expressive patterns to create new content."**

Axel Brando (axel.brando@bsc.es, LinkedIn)

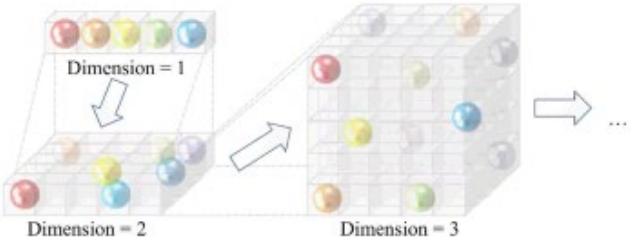
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The Curse of Dimensionality Phenomena

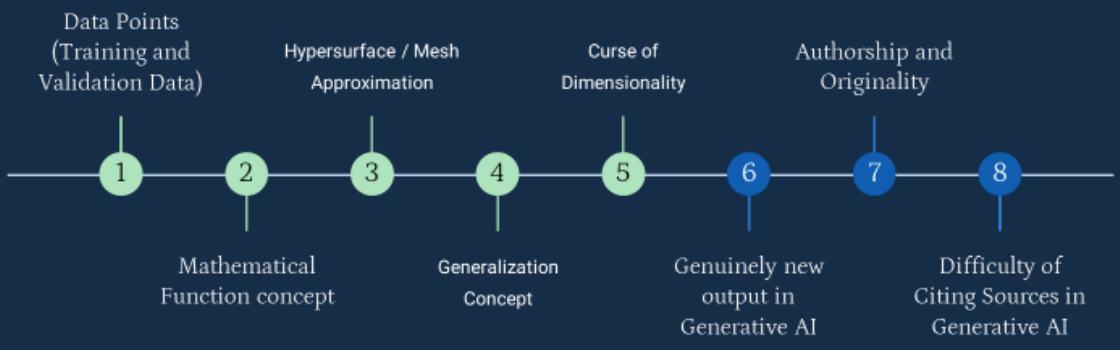
- As the number of **dimensions** increases:
  - The volume of the space grows exponentially.
  - **Distance loses meaning** – everything becomes far apart.
  - Models require vast amounts of data to properly learn.



This exacerbates the problem:

- **We need trillions of points** to approximate the hypersurface well.
- It becomes even harder to know whether a generated output is genuinely "new" or simply an artifact close to an original work.

Key Concepts We Need to Define and Understand

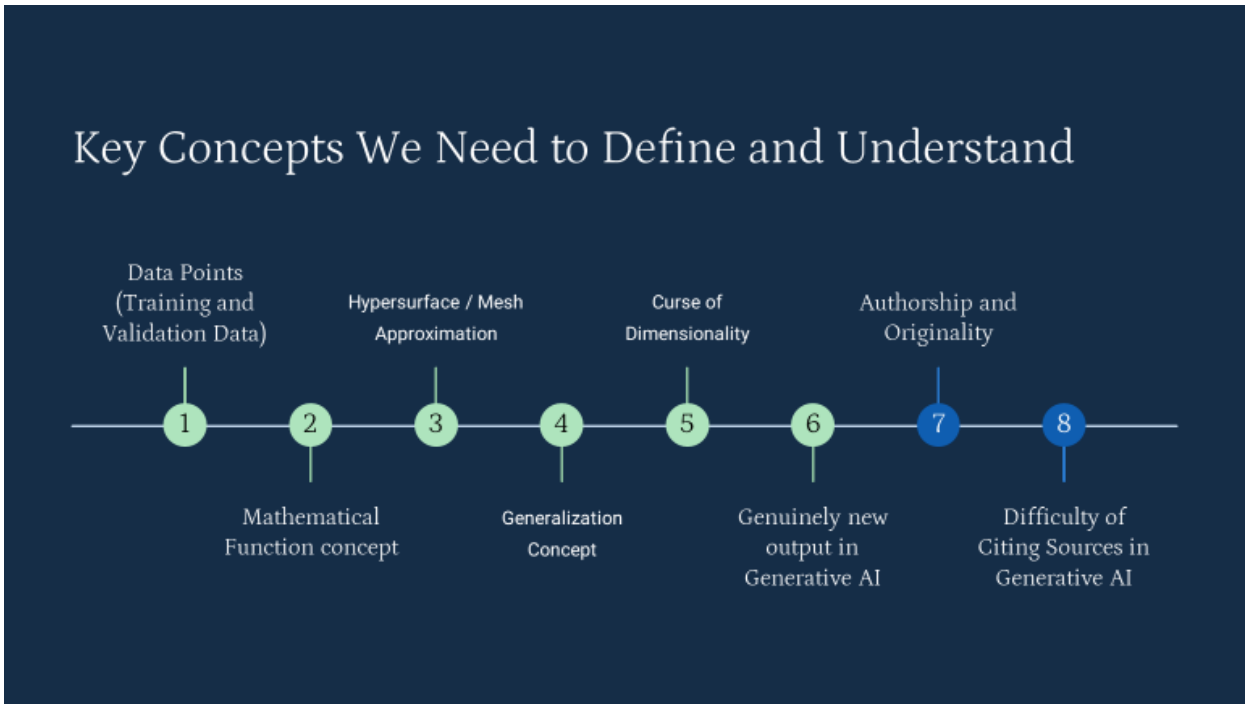


Genuinely "New" Outputs or Stochastic Parroting

- Sometimes, models **stochastically parrot** original data:
  - They reproduce fragments or patterns too close to the original, unintentionally.
  - This happens because the hypersurface, in dense regions, may have been heavily influenced by specific data points.

- Although this is a statistical artifact and not deliberate copying, **distinguishing genuinely new outputs from unintended replicas is extremely difficult at scale.**
- Distinguishing genuinely new (novelty) outputs from unintended replicas is extremely difficult at scale.

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Authorship and Originality

Human Creation

We reinterpret ideas and express them in original ways, with intent and understanding

AI Generation

Models statistically reconstruct patterns learned from data without understanding but with powerful embedding spaces to create new content

Copyright law protects expressions, not ideas.

Since the model learns expressions (patterns of expression), there is a legal risk.

Key challenge: How can we ensure proper *novelty*, as previously discussed, and *attribution*, which we will now examine, in AI-generated works?

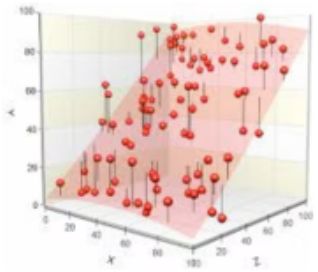
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Key Concepts We Need to Define and Understand

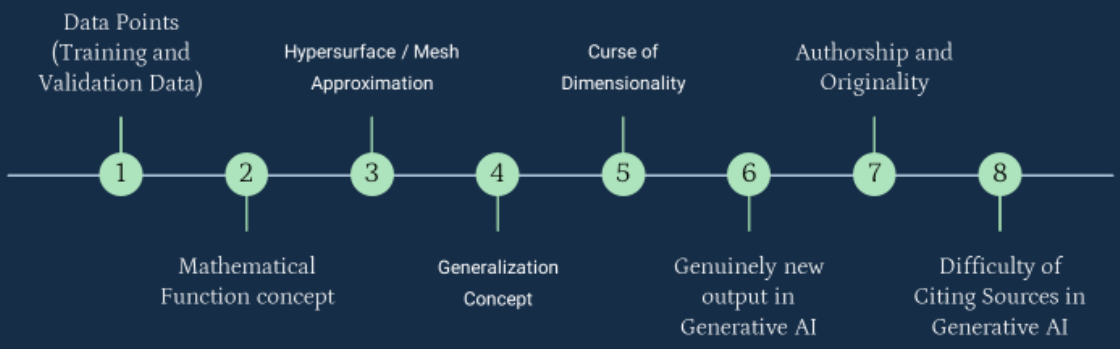
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Difficulty of Citing Sources in AI Outputs (Attribution)

- Due to:
  - **Stochastic training**,
  - **Data not stored** after training,
  - **Massive scale of data**,
- **We cannot trace back** which specific data points influenced a given output.
- It is a research challenge to "cite" sources, as we do in writing (e.g., state of the art section in papers).
- This creates a **traceability gap**:
  - Outputs are "based on" millions or billions of Data Points, so the precise lineage is complex.



Key Concepts We Need to Define and Understand



Conclusions

Hypothesis: "AI models do not learn like humans; instead, they statistically approximate data points through high-dimensional hypersurfaces. New data points are generated based on this learned hypersurface. This fundamental difference has profound implications for authorship and traceability."

1. **We explained step-by-step** how AI models are trained — from the ingestion and statistical modeling of expressive copyrighted works, through hypersurfaces, to the generation of new outputs.
2. **We emphasized** that each step — ingestion, processing, learning — necessarily involves accessing and copying protected content, not merely ideas.
3. **We demonstrated** that the training process relies on constructing a statistical hypersurface rather than memorizing data points, **which complicates traceability and proper attribution**.

Thus, we showed that generative AI models approximate the original data through a high-dimensional hypersurface, leaving open **two challenges**: (1) **Novelty**: verifying whether the generated content is genuinely new, and (2) **Attribution**: tracing them back to the original data sources once the hypersurface has been learned.

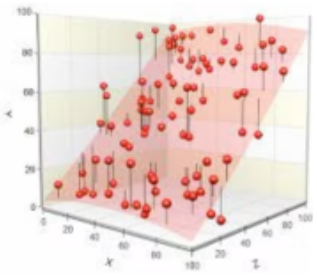
THANK YOU

Axel Brando – axel.brando@bsc.es - linkedin.com/in/axelbrando

European Parliament on Generative AI and copyright Workshop 10

Limitations of Detecting AI-Generated Content (real vs fake content)

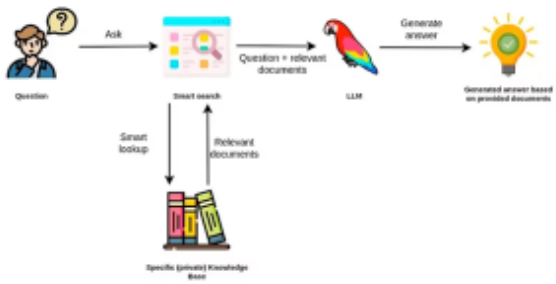
- It's equally difficult to determine whether a piece of content was generated by AI.
- The outputs are plausible and statistically sound.
- **Watermarking** or **fingerprinting** techniques are still under research and not widely reliable.



European Parliament on Generative AI and copyright Workshop – Axel Brando (axel.brando@bsc.es, LinkedIn) A1

RAG: Why It Does Not Solve These Issues

- **Retrieval-Augmented Generation (RAG)**:
  - Instead of training all knowledge into the model, RAG retrieves information from external databases at generation time.
- **Legally**:
  - It is closer to a search engine querying external documents.
  - **It doesn't eliminate the reproduction issue** for models that have already been trained on protected content.
- RAG can reduce risks, but **it does not address the core legal problem** of how the training phase was conducted.



European Parliament on Generative AI and copyright Workshop – Axel Brando (axel.brando@bsc.es, LinkedIn)

Short bio: Axel Brando, PhD

2009 - 2015 — **Degree in Mathematics | Degree in Computer Science**

Dual EHEA Bachelor's Degree in Mathematics and Computer Engineering.

2015 - 2017 — **Researcher at the University of Barcelona (UB)**

Working on style transfer and starting to research on Uncertainty Modelling using Neural Networks.

UNIVERSITAT DE BARCELONA

2015 - 2017 — **Master's Degree in Artificial Intelligence**

Master's thesis about modelling a certain kind of uncertainty using DL. Public repository with 300 stars (https://github.com/axelbrando/Mixture-Density-Networks-for-distribution-and-uncertainty-estimation)

2017 — **Researcher at the National Council IIIA - CSIC**

Applied research in an industrial project.

IIIA Institut d'Investigació en Intel·ligència Artificial — CSIC

2017-2020 — **Industrial Data Scientist Researcher at the bank BBVA**

Industrial researcher inside the Spanish bank BBVA developing research that was connected to real needs of the bank in order to maximize the impact on their applications. This research has been applied to solving internal problems that involve predicting risk such as:

- Impending / Expenses balance forecasting with reliable projection.
- Client movements classifier with confidence.
- Risk inference for several internal problems.
- Several publications in international conferences such as **NeurIPS, AISTATS, ECML-PKDD** and other international journals.

BBVA

The work was part of the **presented Industrial PhD thesis** (done also in the UB) about: Uncertainty Modelling using Deep Learning (2017-2022, Cum laude).

Since then — **Leading the Trustworthy AI research line - BSC group**

The research line is about designing Trustworthy AI-based systems, for problems that require Functional Safety certifications (e.g., autonomous driving, financial and medical forecasters). Inside the BSC my role is to promoting:

- Applied Industrial collaborations (**bilateral projects**) with large/inter- mediate companies.
- Knowledge transfer projects (e.g., R+D+I, Industrial PhDs)
- **European Projects** related to Safety AI (our group leads several).
- National projects and others.

BSC Barcelona Supercomputing Center — Centro Nacional de Supercomputación

A3

7.2.3. Presentation of Dr Christian Peukert

European Parliament – JURI – Workshop Generative AI and copyright
June 4, 2025

Unil.

Generative AI, copyright and economics

Christian Peukert
Associate Professor (Digitization, Innovation and Intellectual Property)
University of Lausanne, Faculty of Business and Economics

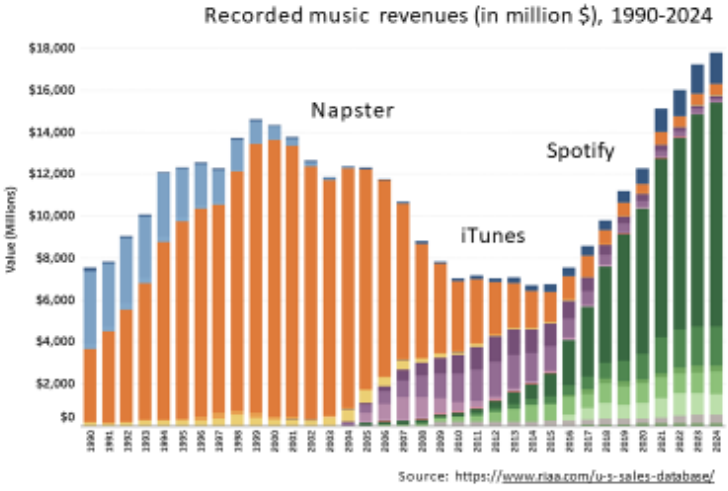
Agenda

- Economic perspective of copyright in the digital age
- Based on 10+ years of own research and review of 160+ academic papers (Peukert and Windisch, 2025) and 10+ papers since then
- Today: historical background & focus on input data
- Conclusion: not enough empirical evidence to propose what to do, but some insights on what not to do

2

Digitization and copyright (1/3)

- Demand-side copyright infringement
  - Fighting online piracy didn't really work
  - But licensing digital services did (iTunes, Spotify, Netflix)



3

European Parliament – JURI – Workshop Generative AI and copyright
June 4, 2025

Unil.

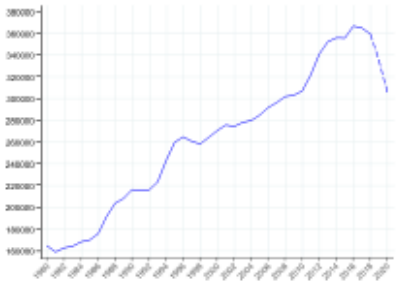
Generative AI, copyright and economics

Christian Peukert
Associate Professor (Digitization, Innovation and Intellectual Property)
University of Lausanne, Faculty of Business and Economics

Digitization and copyright (2/3)

- Digital technology also lowered the cost of creation, distribution, and promotion
- Massive increase in the supply of new creative works
- A large portion is user-generated and may involve the re-use of existing works
- Supply-side copyright infringement
  - Youtube has 14 billion videos of which about 50% include music





4

Digitization and copyright (3/3)

- Shift from pure enforcement (Notice & Takedown) to enabling a digital licensing market for re-use (ContentID)
- Rightsholders can monetize detected uses
- Low-cost licensing of billions of derivative works, substantial revenue for rightsholders
- However:
  - Circumvents traditional institutions involved in copyright settlement
  - Potential for fraud and biases towards economically powerful rightsholders (Arnorsson et al. 2025)

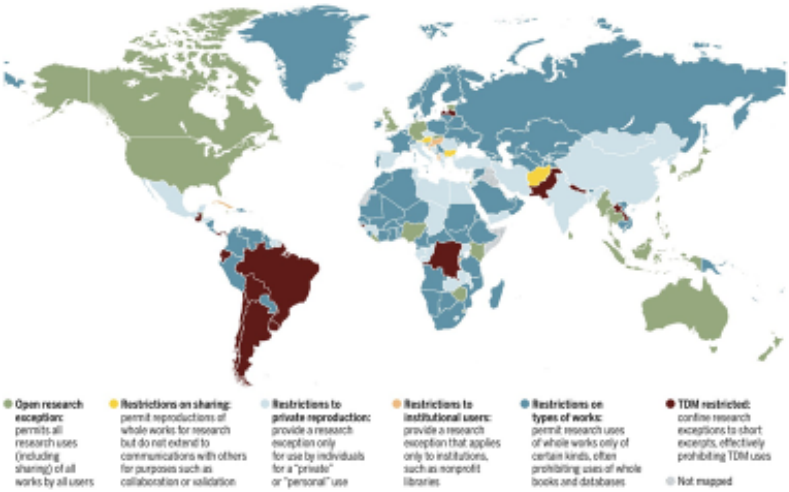
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What is different about GenAI?

	File-sharing	User-Generated-Content	Generative AI
Who loses?	(Most) rightsholders	Possibly rightsholders (but may benefit too)	Possibly rightsholders (but may benefit too)
Who wins?	Consumers	Consumers + New creators	Almost everyone (workers, firms, scientists, creators)
Solution	Licensing (e.g., iTunes & Spotify)	Algorithmic licensing (e.g., ContentID)	?

6

"Outdated copyright laws around the world hinder research" (Fiil-Flynn et al, 2022, Science)



Less restrictive copyright law correlates with

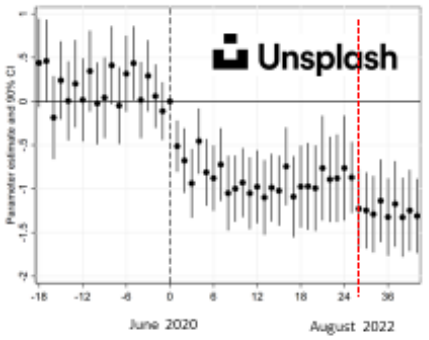
- more AI research output
- more AI code contributions
- more AI patents
- more new AI firms and more investment

(Peukert, 2025)

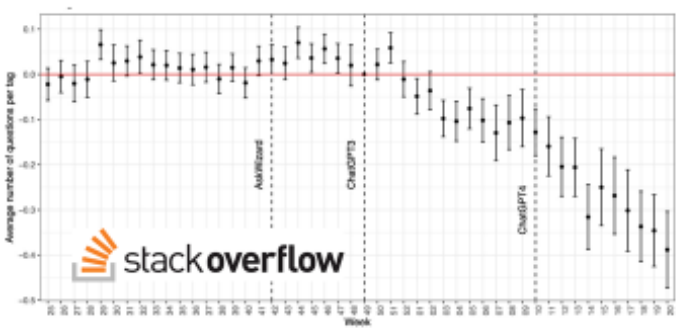
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However, without compensation, will we have enough high-quality data tomorrow?

Peukert et al. (2024)



Quinn and Gutt (2025)



8

What if we let rightsholder decide?

- Audit of LAION (Shcherbakov et al., 2025)
- Largest public image-text dataset for foundation models: 2 billion images ≈ every image on the internet
- Opt-in (0.1%): Images with license information
  - Smaller in size and resolution
  - Very different contents
  - 10 websites account for 66% of all images
- Opt-out (2%): Images registered in 3rd party database
  - Smaller in size and resolution
  - Very different contents
  - 10 websites account for 93% of all images

9

Policy options

(based on theory)

1. Do nothing: Licensing market will develop
   - Example: Google, OpenAI, etc. now have deals with publishers
   - Let the courts decide (taking market power into consideration; Colangelo, 2025)
2. Bet on technology: Algorithmic licensing
   - Compensation = input data's contribution to output (Wang et al., 2024)
   - Risk-management: Predicting opt-in/opt-out (Shcherbakov et al., 2025)
   - Practical challenge: Data provenance is difficult to establish (Longpre et al. 2024)
3. Statutory licensing (e.g. Jaffe, 2025 USCO Report)
   - Economically inefficient: potential misallocation and administrative cost
   - Perhaps only for large foundational models, where direct licensing is close to impossible (Gans, 2025)

10

Copyright policy needs to find a balance

Interests of rightsholders

Research and development



Interests of society

- **Adoption of genAI is 2x faster than previous tech**
  - 40% US population after 2 years (vs 20% for computers & internet) (Bick et al., 2024)
- **Productivity enhancement**
  - AI assistance increases worker productivity by 15% (Brynjolfsson et al., 2025)

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Thank you

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The JURI Committee workshop on generative AI and copyright was held on June 4, 2025. Prof. Dr Lucchi presented a study requested by the Committee on the legal aspects of the phenomenon.The complex relationship between copyright and generative AI from the perspective of technological and economic dimensions was addressed by the accompanying expertise of Dr Axel Brando and Prof. Dr Peukert, respectively.
The workshop included a perspective from the creative sector, as well as a presentation of views from the European Commission.
These proceedings are prepared by the European Parliament's Policy Department for Justice, Civil Liberties and Institutional Affairs for the JURI Committee.

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